19/03/2021

Code-A

Aakash
Medical|IIT-JEE|Foundations
Regd. Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph. 011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - 14

Topics Covered : Complete Syllabus of Class-XI & XII (NEET Pattern)

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.
(ii) Mark should be dark and should completely fill the circle.
(iii) Dark only one circle for each entry.
(iv) Dark the circle in the space provided only.
(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.
(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. Consider the following two statements and choose the correct one :
Statement (1): Everything can be proved in physics.
Statement (2) : An axiom is a truth which is self-evident.
(1) Only statement (1)
(2) Only statement (2)
(3) Both statement (1) and statement (2)
(4) Neither statement (1) nor statement (2)

2. Which set of equipment is most suited to measure an object's gravitational mass?
(1) A spring and a stopwatch
(2) A balance, and a set of known masses
(3) A stopwatch and a cliff to drop mass of
(4) A string and a stopwatch

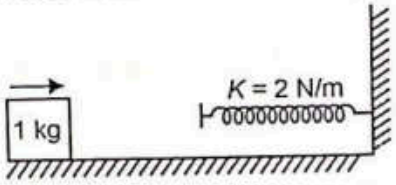
3. The sun's angular diameter is measured to be 1920", The distance of the sun from the earth is 1.4960×10^{11}m. What is the diameter of the sun?
(1) 1.39×10^{9} m (2) 1.39×10^{19} m
(3) 11.39×10^{16} m (4) 2.39×10^{29} m

4. A small mass m suspended from a light inextensible string is given a horizontal speed equal to $\sqrt{2gL}$ where L is the length of the string. The maximum angular displacement of the mass from the vertical will be
(1) 60° (2) 90°
(3) 45° (4) 30°

5. A particle of 1 kg mass, initially suspended from a 50 cm long string vertically, is projected horizontally with a speed 10 m/s. The tension in the string when the particle is at a height 50 cm (from lowest point) is
(1) 10 N (2) 50 N
(3) 100 N (4) 180 N

6. The potential energy of a particle in a certain field has the form $U = a/r^2 - b/r$, where a and b are positive constants, r is the distance from the centre of the field. Find the value of r_0 corresponding to the equilibrium position of the particle.
(1) $\frac{2a}{b}$ (2) $\frac{2b}{a}$
(3) At both (1) and (2) (4) None of these

7. A 1.0 kg block collides with a horizontal light spring of force constant 2 N/m. The block compresses the spring 4 m from the rest position. Assuming that the coefficient of kinetic friction between the block and the horizontal surface is 0.25, what was the speed of the block at the instant of collision?



(1) 8.9 m/s (2) 7.18 m/s
(3) 6.20 m/s (4) 8.2 m/s

8. A torque of value 5 N-m acts on a rigid body, initially at rest, with moment of inertia 5 kg-m^2. The angular velocity of the body after 3 s will be

(1) 4 rad/s (2) 1 rad/s
(3) 3 rad/s (4) 2 rad/s

9. A torque which is a function of time as (6t), acts on a body. The total angular impulse acting on the body in time interval $t = 0$ to $t = 2$ s is

(1) 22 kg-m^2/s (2) 2 kg-m^2/s
(3) 12 kg-m^2/s (4) 10 kg-m^2/s

10. Center of mass of hollow cone(uniform) from base center is nH, where H is the height of cone and base radius is R. Value of n is

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$
(3) $\frac{1}{4}$ (4) None of these

11. A particle of mass 4 kg starts its motion from rest with a constant acceleration 1 m/s^2. In the system there is another particle of same mass which is moving with a constant velocity of 2 m/s. Which among the following is not a possible value of the speed of COM after 2 s?

(1) 2 m/s (2) 4 m/s
(3) 1 m/s (4) 0.5 m/s

12. As we move radially outwards from center of earth, then acceleration due to gravity

(1) Increases
(2) Decrease
(3) Remains same
(4) First increases then decreases

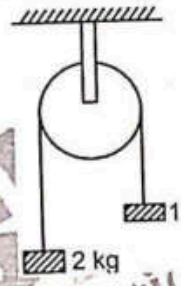
13. The mass of the moon is 7.34×10^{22} kg and the radius is 1.74×10^6 m. The value of gravitational force on unit mass on Moon's surface will be

(1) 1.45 N/kg (2) 1.55 N/kg
(3) 1.75 N/kg (4) 1.62 N/kg

14. Change in elastic modulus for ordinary materials between 0 K and melting point

(1) Increases by 10-20%
(2) Decreases by 10-20%
(3) Remains constant
(4) Increases by 80-90%

15. Two blocks of masses 1 kg and 2 kg are connected by a metal wire going over a smooth pulley as shown. The breaking stress of the metal is $\frac{40}{3\pi} \times 10^6 \text{N m}^{-2}$. If $g = 10$ m s^{-2}, then what should be the minimum radius of the wire used if it is not to break?



(1) 0.5 mm (2) 1 mm
(3) 1.5 mm (4) 2 mm

16. A 4.0 m long copper rod of cross sectional area 1 cm^2 is stretched by a force of 4.0×10^3 N. If Young's modulus for copper is $Y = 1.0 \times 10^{11}$ N/m^2, then stress is

(1) 2.0×10^7 N/m^2
(2) 4.0×10^7 N/m^2
(3) 1.0×10^7 N/m^2
(4) 6.0×10^7 N/m^2

17. If 20 cm and L are the lengths of rods A and B respectively with coefficients of linear expansion 10^{-5} SI units and 2×10^{-5} SI units respectively. The value of L for which the difference in lengths is constant at all temperatures, is

(1) 15 cm
(2) 5 cm
(3) 20 cm
(4) 10 cm

18. As the temperature of water is increased, its density
(1) Decreases
(2) Increases
(3) Remains same
(4) May increase or decrease

19. An engine works between the reservoir temperatures 100 K and 200 K. Which among the following cannot be possible efficiency of the engine?
(1) 46% (2) 15%
(3) 32% (4) 60%

20. Consider the following statements and choose the incorrect one
(1) According to second law of thermodynamics, no engine can have 100% efficiency
(2) Irreversible engines have more efficiency than reversible engines which consist of reversible quasi-static processes
(3) A Carnot engine is an ideal reversible engine that works between two reservoirs at different temperatures
(4) All of these

21. When a particle executes simple harmonic motion without any external effects such as damping,
(1) Kinetic energy does not vary with time
(2) Potential energy remains constant with time
(3) Total energy remains constant
(4) Total energy vary with time

22. A particle of mass 1 kg is performing SHM and its displacement any time is given by

$$y = 2\sin\left(\frac{\pi}{12}t + \frac{\pi}{6}\right)m$$

What will be kinetic energy of particle at $t = 2$ s?
(1) $\frac{\pi^2}{288}$ J
(2) $\frac{\pi^2}{144}$ J
(3) $\frac{\pi}{288}$ J
(4) $\frac{\pi}{144}$ J

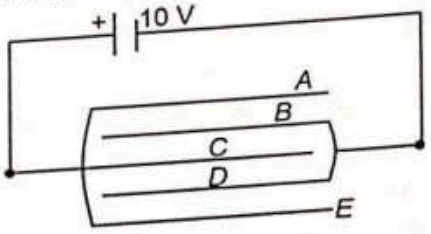
23. Regarding oscillation of the body in damped harmonic oscillations
(1) The amplitude of oscillation of the body decreases with time
(2) The amplitude of oscillation of the body increases with time
(3) The amplitude of oscillation of the body remains constant with time
(4) None of these

24. Energy of damped oscillator becomes half of its initial value (E_0) in 1 sec Energy of damped oscillator after 3 sec will be
(1) $\frac{E_0}{4}$ (2) $\frac{E_0}{8}$
(3) $\frac{E_0}{9}$ (4) $\frac{E_0}{10}$

25. A proton is released from rest, 10 cm from a charged sheet carrying charge density of -2.21×10^{-9} C/m^2. It will strike the sheet after the time (approximately)
(1) 4 μs (2) 2 μs
(3) $\sqrt{2}$ μs (4) $4\sqrt{2}$ μs

26. Electric field due to conducting parallel plates at any point near to a plate is E then what is the electric potential difference on moving a distance x parallel to the plates
(1) Zero (2) $2Ex$
(3) $Ex/2$ (4) Ex

27. Five identical capacitor plates are arranged such that they make four capacitors each of 2 μF. The plates are connected to a source of emf 10 V. The charge on plate C is



(1) +20 μC
(2) +40 μC
(3) +60 μC
(4) +80 μC

28. The charge passing through a conducting wire depends on time as $q = 4\cos 2t$. Then find the maximum value of instantaneous current.

(1) 1 A (2) 8 A

(3) 2 A (4) 4 A

29. The charge flowing through resistor R varies with time t as $Q = at^2 - bt^3$. where a and b are positive constants. Find the time when current is maximum or minimum.

(1) $2a/3b$ (2) $2a/b$

(3) $a/3b$ (4) $a/6b$

30. Consider a current carrying straight wire, carrying a current along positive x direction. If magnetic field exists in the region in the direction along negative x direction. The force acting on the wire due to the magnetic field is

(1) Along z direction

(2) Along y direction

(3) Along x direction

(4) Zero

31. Among following two statements, choose the correct one(s).

Statement (1) : If magnetic force is non-zero then it is always perpendicular to the velocity of the moving charged particle.

Statement (2) : Magnetic force may be zero on a moving charged particle.

(1) Only statement (1)

(2) Only statement (2)

(3) Both statement (1) and statement (2)

(4) Neither statement (1) nor statement (2)

32. The graph between the magnitude of magnetic force as a function of the speed of a charged particle, when it is projected perpendicularly to a given uniform magnetic field, is

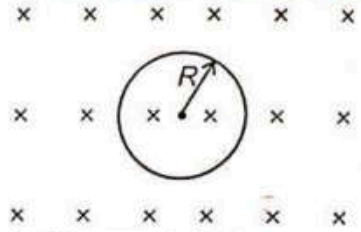
(1) Straight line parallel to speed axis

(2) Straight line, inclined at an angle to speed axis

(3) Hyperbola

(4) Parabola

33. A circular coil of radius 'R' is placed in variable magnetic field as shown in figure.

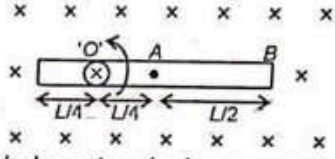


If strength of magnetic field as a function of time is $5t^2$ and resistance of the coil is $\gamma\ \Omega$ then the induced current in the coil at any instant is

(1) $10\pi\gamma R^2 t$ (2) $\dfrac{5t^2R}{\gamma}$

(3) $\left(\dfrac{5\pi R^2}{\gamma}\right)t$ (4) $\left(\dfrac{10\pi R^2}{\gamma}\right)t$

34. A conducting rod is rotating about a fixed point as shown in the figure with angular velocity ω



If the whole setup is in perpendicular uniform magnetic field B_0, then the potential difference between point 'A' and 'B' $(V_A - V_B)$ is

(1) $-\dfrac{B_0\omega L^2}{2}$ (2) $-\dfrac{B_0\omega L^2}{4}$

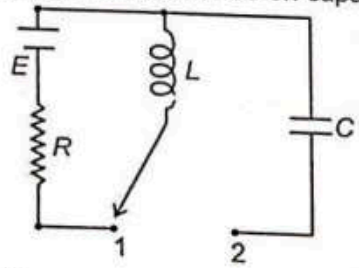
(3) $\dfrac{B_0\omega L^2}{2}$ (4) $\dfrac{B_0\omega L^2}{4}$

35. The charge on the capacitor in LC circuit is given by equation $\dfrac{d^2Q}{dt^2} + 16Q = 0$. Find the frequency of LC oscillation?

(1) $2/\pi$ Hz (2) 2π Hz

(3) 4 Hz (4) $4/\pi$ Hz

36. Switch is in position 1 for a long time. At time $t = 0$, it is shifted to position 2. Find the maximum charge which will accumulate on capacitor.

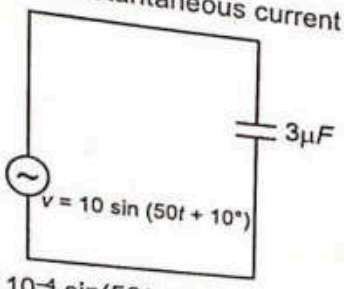


(1) $\dfrac{E}{R\sqrt{LC}}$ (2) $\sqrt{LC}\left(\dfrac{R}{E}\right)$

(3) $\sqrt{LC}\left(\dfrac{E}{R}\right)$ (4) $\dfrac{R}{E\sqrt{LC}}$

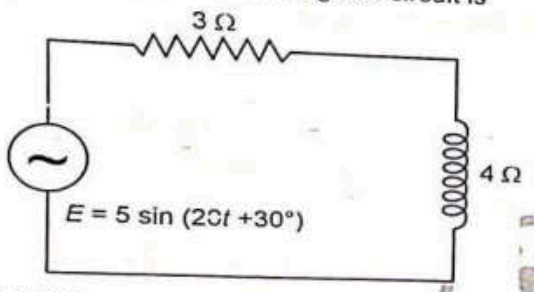
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37. For the circuit shown in the figure. The expression of instantaneous current is



(1) $15 \times 10^{-4} \sin(50t - 80°)$
(2) $15 \times 10^{-4} \sin(50t + 100°)$
(3) $1.5 \times 10^{-4} \sin(50t + 100°)$
(4) $1.5 \times 10^{-4} \sin(50t - 80°)$

38. The net impedance of the given circuit is



(1) 3 Ω
(2) 4 Ω
(3) 5 Ω
(4) 7 Ω

39. In an electromagnetic wave E = (1 V/m) sin ($wt - kx$). The average energy density of magnetic field, in SI units will be
(1) 1.1×10^{-11}
(2) 2.2×10^{-12}
(3) 3.3×10^{-13}
(4) 4.4×10^{-14}

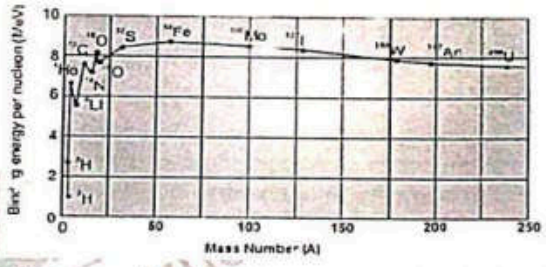
40. If frequency of an EMW is 5×10^{14} Hz, then the number of waves in 1 mm length in a medium of refractive index 1.5 will be
(1) 2.5 (2) 25
(3) 250 (4) 2500

41. Dispersive power of prism is defined as the ratio of angular dispersion to
(1) Mean deviation
(2) Refractive index
(3) Wavelength of red light
(4) Speed of light

42. A wave of wavelength 400 nm superimposes with another wave (coherent with first) of same wavelength. If the path difference between the waves is 300 nm, then the resultant intensity due to superposition will be
(1) Maximum
(2) Minimum
(3) Between minimum and maximum
(4) Half of maximum wavelength

43. A particle A has momentum P, while another particle B has a momentum $2P$. The momentum of the particles increase with time but their ratio always remains 1 : 2. The de-Broglie wavelength of the particles (that of A to that of B)
(1) Increases with time
(2) Decreases with time
(3) Remains same and equal to 1 : 2
(4) Remains same and equal to 2 : 1

44. Among the following statements choose the incorrect one regarding the following plot



(1) For nuclei of middle mass number (30 < A < 170) the binding energy per nucleon is almost constant
(2) For both light nuclei (A < 30) and heavy nuclei (A > 170) binding energy per nucleon is lower
(3) There will be a loss in overall binding energy when we move from a heavy nuclei region to the middle region of the plot
(4) All of these

45. In which of the following configuration both voltage gain and current gain are higher than unity?
(1) Common emitter configuration
(2) Common base configuration
(3) Both (1) and (2)
(4) None of the above

CHEMISTRY

46. Molality of an aqueous solution of NaOH in which mole fraction of NaOH is 0.25

(1) 10.5 m (2) 18.5 m

(3) 5 m (4) 20.2 m

47. Dilute aqueous solution of boric acid is used as

(1) Tranquilizer (2) Analgesics

(3) Antiseptics (4) Antifertility drug

48. The wave number of a radiation of wavelength of 5000Å is

(1) 200 cm^{-1} (2) 2000 cm^{-1}

(3) 20000 cm^{-1} (4) 40000 cm^{-1}

49. Incorrect value of quantum number for last electron of magnesium is

(1) n = 3 (2) $l = 0$

(3) m = – 1 (4) $s = +\frac{1}{2}$

50. Select the element of highest negative electron gain enthalpy among the following

(1) C (2) F

(3) S (4) P

51. In carius method of estimation of halogen, 0.35 g of an organic compound gave 0.14 g of AgCl then the percentage of chlorine in the compound is

(1) 48.24% (2) 34.51%

(3) 9.89% (4) 42.82%

52. Select the correct statement among the following

(1) Glycine is achiral amino acid

(2) Hydrolysis product of lactose is known as inverted sugar

(3) Low density polyethene is a linear chain polymer

(4) Nylon 6, 6 is addition polymer of urea and formaldehyde

53. Excess amount of which chemical in drinking water mainly damages kidney, brain and liver.

(1) Nitrates (2) Fluorides

(3) Lead (4) Sulphates

54. Pair of species of same magnetic property is

(1) O_2, N_2 (2) O_2, C_2^{2-}

(3) N_2^+, N_2^- (4) C_2, H_2^-

55. Select the molecule in which all atoms are lying in same plane?

(1) $CHCl_3$ (2) XeF_4

(3) PCl_3F_2 (4) XeO_4

56. Among the following, strongest hydrogen bond is present in

(1) F — H - - - O (2) N — H - - - O

(3) O — H - - - F (4) O — H - - - N

57. Which species has highest ionization energy?

(1) Na (2) Mg

(3) Na^+ (4) Mg^{2+}

58. Density of $SO_2(g)$ (in g/L) at 1 atm pressure and at 127°C temperature, is

(1) 1.2 (2) 1.95

(3) 3.95 (4) 10.5

59. If enthalpy of neutralization of a weak monobasic acid (HA) with NaOH is – 7.7 kcal eqv^{-1} then ionization enthalpy of HA is (Enthalpy of neutralization of HCl with NaOH is – 13.7 kcal eqv^{-1})

(1) 21.4 kcal eqv^{-1} (2) 6 kcal eqv^{-1}

(3) 19.7 kcal eqv^{-1} (4) 12 kcal eqv^{-1}

60. For isothermal reversible expansion of 2 mol ideal gas. Which is necessarily true?

(1) W > 0 (2) $\Delta U < 0$

(3) $\Delta S_{surr} = 0$ (4) Q > 0

61. Intensive property among the following is

(1) Specific heat capacity

(2) Heat capacity

(3) Resistance

(4) Mole

62. In 10 L vessel, 2 mol PCl_5 decomposes into PCl_3 and Cl_2 at 127°C. If at equilibrium 40% PCl_5 decomposes then Kc of reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, is

(1) $\frac{16}{75}$ (2) $\frac{4}{75}$

(3) $\frac{1}{75}$ (4) $\frac{1}{150}$

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63. If 1 L aqueous solution of 0.05 M H_2SO_4 is diluted to 1000 L then pH of the final solution is
(1) 2
(2) 3
(3) 4
(4) 3.5

64. Which one of the following pair of solution form an acidic buffer?
(1) H_3PO_4 and NaH_2PO_4
(2) Na_2S and NaOH
(3) HNO_3 and H_2SO_4
(4) CH_3COOH and HCN

65. Which one of the following is most soluble?
(1) Ag_2CO_3 ($K_{sp} = 8 \times 10^{-12}$)
(2) $CaCO_3$ ($K_{sp} = 3 \times 10^{-9}$)
(3) $CuCO_3$ ($K_{sp} = 1.4 \times 10^{-10}$)
(4) $PbCO_3$ ($K_{sp} = 7.4 \times 10^{-14}$)

66. Select the reaction in which H_2O_2 is acting as reducing agent?
(1) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$
(2) $I_2 + H_2O_2 + 2OH^- \longrightarrow 2I^- + 2H_2O + O_2$
(3) $2Fe^{2+} + 2H^+ + H_2O_2 \longrightarrow 2Fe^{3+} + 2H_2O$
(4) $Mn^{2+} + H_2O_2 \longrightarrow Mn^{4+} + 2OH^-$

67. Standard reduction potentials for Pb^{2+} / Pb and Au^{3+}/Au couples respectively are – 0.13 V and 1.4 V. These two couples in their standard state are connected to make a cell. The cell potential of the cell will be
(1) 1.27 V
(2) 1.53 V
(3) 2.93 V
(4) 2.8 V

68. Percentage strength of 16.8 volume solution of hydrogen peroxide is
(1) 3 (2) 4
(3) 5.1 (4) 10.2

69. Potassium on reaction with excess of air gives
(1) K_2O (2) KO_3
(3) KO_2 (4) K_2O_2

70. Select the incorrect statement about similarities between lithium and magnesium.
(1) Chlorides of both are soluble in ethanol
(2) Chlorides of both are deliquescent
(3) Chlorides of both hydrated with two water molecules
(4) Both elements form nitrides with nitrogen present in air

71. Aldol condensation will not take place in
(1) CH_3CHO (2) CH_3CH_2CHO
(3) $(CH_3)_2CHCHO$ (4) $PhCH_2CHO$

72. Which amine can be prepared by Gabriel phthalimide process?
(1) $CH_3CH_2NH_2$ (2) $PhNHCH_3$
(3) $PhCH_2NHCH_3$ (4) $(CH_3)_3N$

73. Thermodynamically most stable allotrope of carbon is/are
(1) Diamond
(2) Graphite
(3) Fullerene
(4) All are equally stable

74. Consider the following reactions sequence
$B_2H_6 + NH_3 \longrightarrow$ A (Ionic compound) $\xrightarrow{\Delta}$ B + $H_2(g)$. A and B respectively are
(1) H_3BO_3 and BN
(2) $[BH_2(NH_3)_2]^+ [BH_4]^-$ and BN
(3) $[BH_2(NH_3)_2]^+ [BH_4]^-$ and $B_3N_3H_6$
(4) $[BH_2(NH_3)_2]^+ [BH_4]^-$ and B_2O_3

75. If 0.2 molal aqueous solution of K_3PO_4 is 95% ionized then the freezing point of this solution will be ($K_f(H_2O)$ = 1.86 K molal^{-1})
(1) 1.43 K (2) 271.57 K
(3) 274.43 K (4) 270.14 K

76. An unknown solid [X] exhibits fcc structure at room temperature. At higher temperature T(°C), it transforms to simple cubic structure. The ratio of density of X at room temperature to that at T (°C) (Assuming molar mass and atomic radii of X remains constant with temperature) is
(1) 1 : 1 (2) $\sqrt{2}$: 1
(3) 1 : 3 (4) 2 : 3

77. Reduction potential of hydrogen electrode at pH = 0 is
(1) – 0.0591 V (2) 0.0591 V
(3) Zero (4) 0.591 V

78. On electrolysis of aqueous solution of $CuCl_2$ (using platinum electrodes) products of cathode and anode respectively are
(1) Cu(s), H_2(g) (2) Cu(s), Cl_2(g)
(3) H_2(g), Cu(s) (4) O_2(g), H_2(g)

79. Consider a first order reaction, P → Q . If the time taken for 1.2 mole of P to produce 0.9 mole of Q is 30 min then time taken for conversion of 1.8 mole of P to produce 1.35 mol of Q will be
(1) 30 min (2) 60 min
(3) 120 min (4) 15 min

80. For the coagulation of methylene blue sol, which ion will have maximum coagulating power?
(1) Ca^{2+} (2) Al^{3+}
(3) SO_4^{2-} (4) $[Fe(CN)_6]^{3-}$

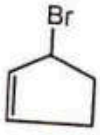
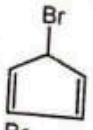
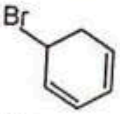
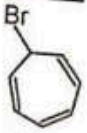
81. Zirconium is refined by
(1) Distillation
(2) Liquation
(3) Electrolytic refining
(4) Vapour phase refining

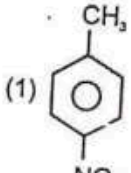
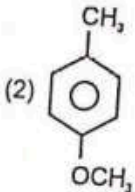
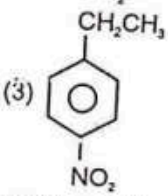
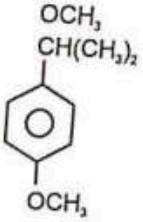
82. Incorrect match of catalyst used in given process is
(1) Deacon's process: $CuCl_2$
(2) Contact process: NO
(3) Ostwald process: Pt/Rh gauge
(4) Haber's process: Iron oxide

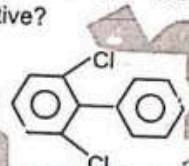
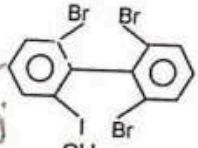
83. Compound of xenon formed in product when XeF_4 will react with O_2F_2
(1) XeF_2 (2) $XeOF_4$
(3) XeO_2F_2 (4) XeF_6

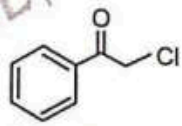
84. Number of unpaired electron(s) present in Yb^{3+} ion is (Atomic number of Yb is 70)
(1) 3 (2) 7
(3) Zero (4) 1

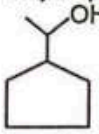
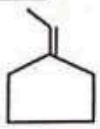
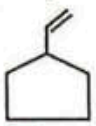
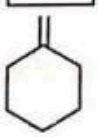
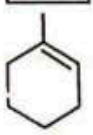
85. Correct electronic configuration of metal ion in $K[Co(CN)_4(NH_3)_2]$ is (atomic number of cobalt is 27)
(1) $t_{2g}^6 e_g^0$ (2) $t_{2g}^3 e_g^3$
(3) $t_{2g}^2 e_g^4$ (4) $t_{2g}^4 e_g^2$

86. Among the following which bromide will react fastest with $AgNO_3$ solution?
(1) 
(2) 
(3) 
(4) 

87. Which among the following give fastest rate of electrophilic aromatic substitution?
(1) 
(2) 
(3) 
(4) 

88. Which of the following molecule is optically active?
(1) 
(2) 
(3) Me — CH(CH3) — Cl (4) Et — CH(CH3) — Cl

89. Which one is most reactive towards S_N2 reaction?
(1) 
(2) CH_3CH_2Cl
(3) CH_3Cl (4) 

90. Major product obtained on acidic dehydration of  , is
(1) 
(2) 
(3) 
(4) 

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BOTANY

91. Which of the following pair of plants show pollination by abiotic agents?
(1) *Yucca* and Maize
(2) *Hydrilla* and *Zostera*
(3) Figs and *Yucca*
(4) *Amorphophallus* and *Eichhornia*

92. The endosperm is completely consumed by the developing embryo in
(1) Pea and beans
(2) Castor and ground nut
(3) Coconut and castor
(4) Pea and coconut

93. Which term is applicable to the offspring formed by vegetative reproduction?
(1) Bud
(2) Vegetative propagule
(3) Clone
(4) Hybrid

94. Apomicts have several advantages in horticulture and agriculture **except**
(1) Maintain hybrid characters in generations
(2) Farmers can keep on using hybrid seeds for new crop
(3) Segregation of characters in hybrid progeny will increase
(4) Progeny can maintain hybrid character

95. The genetic code with dual function is
(1) AUG
(2) UUU
(3) UAA
(4) UAG

96. How many phenotypes and genotypes are possible in the F_2-generation of a typical Mendelian dihybrid cross?
(1) 4 phenotypes and 6 genotypes
(2) 9 phenotypes and 4 genotypes
(3) 4 phenotypes and 9 genotypes
(4) 6 phenotypes and 6 genotypes

97. The female gametophytic plant bodies have an independent free living existence in
(1) *Cycas* (2) *Marchantia*
(3) *Cedrus* (4) *Dryopteris*

98. Select the option with **correct** match.
(1) Wildlife sanctuary – *Ex-situ* conservation
(2) Biodiversity hotspots – High degree of endemism
(3) Zoological park of India – *In-situ* conservation
(4) Sacred groves of Meghalaya – *Ex-situ* conservation

99. Effect of dormancy promoting substance in seeds can be removed by subjecting them to
(1) Nitrate and gibberellic acid
(2) Nitrate and ascorbic acid
(3) Phenolic acid and gibberellic acid
(4) Para-ascorbic acid and dormin

100. Viruses differ from viroids in having
(1) RNA molecules without protein coat
(2) DNA molecules without protein coat
(3) Both DNA and RNA molecules with protein coat
(4) DNA or RNA molecules with protein coat

101. Which cells of anther secrete sporopollenin?
(1) Sporogenous tissue
(2) Endothecium
(3) Tapetum
(4) Microspore mother cells

102. The carboxylation activity of RuBisCO in chloroplast of C_4-plants leads to the formation of
(1) Phosphoenol pyruvate
(2) Oxalo-acetic acid
(3) Phosphoglyceric acid
(4) Malic acid

103. Which of the following would cause inflammation of cornea, called snow-blindness?
(1) High dose of UV-B radiation
(2) Healing of ozone layer
(3) Global warming
(4) Enhanced greenhouse gases

104. The pyramid of biomass in sea is generally
(1) Upright (2) Urn-shape
(3) Inverted (4) Irregular

105. The unequivocal proof that DNA is the genetic material came from the experiments of
(1) M. Chase and Hershey
(2) Mac Leod and Mc Carty
(3) Watson and Crick
(4) Meselson and Stahl

106. The cross where the F_1 resembles both the parents is
(1) Dominance
(2) Incomplete dominance
(3) Dihybrid cross performed by Mendel
(4) Co-dominance

107. Match the following organisms and their types of sex determination :

	Column-I		Column-II
a.	Human beings	(i)	Female heterogamety
b.	Birds	(ii)	XY-type
c.	Grasshopper	(iii)	Haplo-diploidy
d.	Honey bee	(iv)	XO-type

Select the **correct** option.
(1) a(i), b(ii), c(iii), d(iv)
(2) a(ii), b(i), c(iv), d(iii)
(3) a(iv), b(iii), c(ii), d(i)
(4) a(iii), b(i), c(ii), d(iv)

108. Name the class of fungi which is commonly called imperfect fungi?
(1) Phycomycetes
(2) Ascomycetes
(3) Deuteromycetes
(4) Basidiomycetes

109. In flowering plants, during embryo development the growth is
(1) Initially geometric and later on arithmetic
(2) Initially arithmetic and later on geometric
(3) Geometric throughout
(4) Arithmetic throughout

110. The carbohydrates, fatty acid and protein would be broken into A before the entry in aerobic respiratory pathway when these are used as respiratory substrates. 'A' represents
(1) Pyruvic acid
(2) Succinyl Co-A
(3) Acetyl Co-A
(4) α-ketoglutaric acid

111. Study the following data showing the osmotic potential and pressure potential of A, B, C cells of mesophyll of monocot leaf.

		ψ_S(MPa)	ψ_P(MPa)
(i)	A	–0.6	0.2
(ii)	B	–0.5	0.3
(iii)	C	–0.7	0.1

Identify the **correct** option showing direction of movement of water.
(1) A→B→C (2) B→C←A
(3) B←C→A (4) B→C↔A

112. Match the items in column I with those in column II w.r.t. population interactions.

	Column-I		Column-II
a.	Orchid-Mango tree	(i)	Predation
b.	Fungi-Roots of higher plants	(ii)	Amensalism
c.	*Penicillium*-Bacteria	(iii)	Mutualism
d.	Sparrow-Seeds	(iv)	Commensaiism

Select the **correct** option.
(1) a(i), b(ii), c(iv), d(iii)
(2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(i), c(ii), d(iii)
(4) a(iv), b(iii), c(ii), d(i)

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113. Which of the following elements helps in the formation of middle lamella of plant cells?
(1) Mg and Mn (2) Ca and Mg
(3) Mo and Mn (4) Ca and Zn

114. The biosynthesis of steroidal hormones occurs in
(1) SER (2) RER
(3) Golgi complex (4) Centrosome

115. The decondensation of chromosomes occurs in all phases, **except**
(1) Telophase (2) Telophase II
(3) Telophase I (4) Leptotene

116. The cells of which region of the root-tip form delicate, thread like structures meant for absorption of water and minerals?
(1) Region of root cap
(2) Region of elongation
(3) Region of maturation
(4) Region of meristematic activity

117. Select the **odd** one w.r.t. the pitcher plant and venus-fly trap plant?
(1) Insectivorous (2) Photosynthetic
(3) Saprophytic (4) Flowering plants

118. The mesophyll is not differentiated into palisade and spongy parenchyma in plants having
(1) Reticulate venation in leaf
(2) Parallel venation in leaf
(3) Stomata mainly on lower surface of leaf
(4) Conjoint, collateral and open vascular bundles in leaf

119. Choose the **incorrect** statement w.r.t. activity of the cambium ring in dicot stem.
(1) Primary phloem gets crushed
(2) Secondary phloem gets crushed
(3) Primary xylem get crushed and pushed to periphery
(4) Secondary medullary rays are formed

120. The vexillary aestivation of papilionaceous corolla is represented in floral formula, as
(1) $A_{(9)+1}$ (2) $P_{(3+3)}$
(3) $C_{1+(2)+(2)}$ (4) $C_{1+2+(2)}$

121. The pollen tube enters through the micropyle of
(1) Dicotyledonous seed
(2) Monocotyledonous seed
(3) Megasporangium
(4) Microgametophyte

122. Flowers can be divided into two similar halves by any vertical plane passing through the centre, in all pairs, **except**
(1) *Cassia* and canna
(2) *Datura* and tomato
(3) Chiili and potato
(4) Belladonna and *Colchicum*

123. Identify the name and location of tissues having cells that divide all their life in higher plants.
(1) Parenchymatous tissue at root-tip
(2) Meristematic tissue at shoot-tip
(3) Meristematic tissue of axillary bud
(4) Meristematic tissue of leaf primordium

124. In alcoholic fermentation
(1) Electrons are accepted by acetaldehyde formed by decarboxylation of pyruvate
(2) Electrons are accepted by ethyl alcohol
(3) TPP and Mn^{2+} are required
(4) NADPH is oxidized to $NADP^+$

125. Which of the following is **correctly** matched w.r.t. microbe and product?
(1) *Penicillium notatum* – Citric acid
(2) *Aspergillus niger* – Butyric acid
(3) *Monascus purpureus* – Statins
(4) *Methanobacterium* – CNG

126. The semi-dwarf wheat varieties were developed at
(1) International centre of wheat and maize improvement, in Delhi, India
(2) IRRI, Philippines
(3) IRRI, Cuttack, India
(4) International centre for wheat and maize improvement in Mexico

127. In DNA replication, the helicase activity requires energy in the form of ATP because the
(1) Hydrogen bonding affinity of complementary strands is very low
(2) Two strands bind together by hydrogen bonds
(3) Nicking and sealing is performed by the enzyme
(4) RNA primers cannot be removed without it

128. In eukaryotes, the introns of hnRNA, are removed and exons are joined in a well defined order during post transcription phase by the process known as
(1) Slicing (2) Splicing
(3) Tailing (4) Capping

129. Among the following statements how many is/are **wrong**?
(a) In the first phase of translation, charged tRNA is formed.
(b) UTR enhances the transcription.
(c) Regulation of *lac*-operon by repressor is referred as positive regulation.
(d) The secondary structure of tRNA looks like a clover-leaf.
(1) Four (2) Three
(3) Two (4) One

130. Which of the following match is **not correct**?
(1) Expressed sequence tags – Genes that expressed as RNA
(2) Sigma factor – A small part of bacterial RNA
(3) Exons – Coding sequences
(4) Negative and positive regulation – Lac operon of *E. coli*

131. Find **odd** one w.r.t. taxon.
(1) Canidae (2) Carnivora
(3) Diptera (4) Primata

132. The life cycle of *Polysiphonia* and *Ectocarpus* are
(1) Haplontic type
(2) Diplontic type
(3) Haplo-diplontic type
(4) Haplontic and diplontic type respectively

133. Pinnately compound leaves are found in
(1) *Riccia* (2) *Pinus*
(3) *Cycas* (4) *Equisetum*

134. In given plants list

Psilotum, *Selaginella*, *Polytrichum*, *Salvia*, *Lycopodium*, *Equisetum*, *Dryopteris*, *Pteris* and *Adiantum*

How many belong to class Pteropsida of Pteridophytes?
(1) Seven (2) Five
(3) Two (4) Three

135. Which of the following is expected to occur in anaphase?
(1) Sister chromatids splits from the centromeric point
(2) Sister chromatids do not separate
(3) Chromosomes undergo condensation
(4) Chromosomes break into pieces

ZOOLOGY

136. Skeletal muscle fibres are
(1) Striated, involuntary and uninucleated
(2) Voluntary, striated and multinucleated
(3) Uninucleated, voluntary and striated
(4) Multinucleated, involuntary and non-striated

137. Gap junctions are present in all, **except**
(1) Synapse (2) Smooth muscle
(3) Cardiac muscle (4) Skeletal muscle

138. Complete the analogy w.r.t. number of carbon atoms and double bond(s) present
Palmitic acid : 16C : 0 : : Arachidonic acid : ____
(1) 20C : 0 (2) 20C : 4
(3) 20C : 1 (4) 18C : 0

139. Among the following, aromatic amino acid is
(1) Serine (2) Alanine
(3) Tyrosine (4) Glutamic acid

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140. Inhibition of succinic dehydrogenase by _____ which closely resembles the substrate _____ in structure, is an example of _____.

Choose the option that fills the blanks **correctly** in the given statement.

(1) Oxaloacetate, malonate, competitive inhibition

(2) Malonate, oxaloacetate, non-competitive inhibition

(3) Malonate, succinate, competitive inhibition

(4) Succinate, oxaloacetate, non-competitive inhibition

141. Which of the following is **incorrect** w.r.t. gastric glands?

(1) Mucus neck cells – Mucus

(2) Peptic cells – Pepsinogen

(3) Parietal cells – HCl

(4) Paneth cells – Alkaline fluid

142. Gross calorific value of proteins is

(1) More than physiological value of carbohydrates

(2) Less than gross calorific value of carbohydrate

(3) Less than physiological value of proteins

(4) More than physiological value of fats

143. Choose the **incorrect** match w.r.t. respiratory volumes.

(1) FRC = ERV + RV

(2) EC = TV + ERV

(3) VC = ERV + TV + IRV

(4) TLC = ERV + TV + IRV

144. Fall in which of the following factors causes right shift in oxygen haemoglobin dissociation curve?

(1) pO_2 (2) H^+ concentration

(3) BPG (4) Temperature

145. During isovolumetric systole,

(1) Blood flows from right ventricle to aorta through semilunar valves

(2) Blood flows from left ventricle to pulmonary artery through bicuspid valves

(3) Volume of blood and pressure in ventricles remains constant

(4) Volume of blood remains same while the pressure increases in ventricles

146. During cardiac cycle, each ventricle pumps out approximately 70 mL of blood in a healthy man with 70 beats per minute. The cardiac output is

(1) 5040 mL / min (2) 4900 mL / min

(3) 9800 mL / min (4) 1080 mL / min

147. ABO blood grouping is the most popularly known blood group. It is called ABO because 'A' in it refers to having

(1) Antigen B on RBCs surface while anti-A antibodies in plasma

(2) No antigen A or B on RBCs surface

(3) No antibodies anti-A or anti-B in plasma

(4) Antigen A on RBCs surface and anti-B antibodies in plasma

148. JGA is a special sensitive region formed by cellular modifications in the **A** and the **B** at the location of their contact.

Choose the option that fill the blanks **correctly**.

	A	B
(1)	PCT	Afferent arteriole
(2)	DCT	Efferent arteriole
(3)	DCT	Afferent arteriole
(4)	PCT	Glomerulus

149. If a person is suffering from dehydration, which of the following will cause restoration of blood volume and GFR?

(1) Release of renin from JG cells

(2) Release of a powerful vasodilator Angiotensin – II

(3) Release of aldosterone directly triggered by Angiotensinogen

(4) Release of ADH from JG apparatus

150. Select the **incorrect** match w.r.t. human skeleton.

(1) Lacrimal bones - 2

(2) Floating ribs - 2

(3) Palm bones - 10

(4) Ankle bones - 14

151. Joint present between radius and ulna just below the elbow is also present between

(1) Femur and tibia

(2) Atlas and axis

(3) Carpal and metacarpal of thumb

(4) Radius and ulna near carpals

152. **Incorrect** statement w.r.t. resting axonal membrane potential is

(1) Axonal membrane is comparatively more permeable to potassium ions

(2) Axonal membrane is nearly impermeable to sodium ions

(3) Axoplasm contains high concentration of K^+ and low concentration of Na^+

(4) Active transport of $2Na^+$ outwards to ECF and $3K^+$ into axoplasm

153. Read the following statements carefully and choose the **correct** option.

A: The cerebellum integrates information received from the semicircular canals of the ear and the auditory system.

B: The midbrain receives and integrates visual, tactile and auditory inputs.

(1) Both the statements are incorrect

(2) Both the statements are correct

(3) Only statement A is correct

(4) Only statement B is correct

154. Read the given statements (A to D) and choose the **correct** option that represents them as True (T) or False (F).

A : A pair of broad bean – sized organs, called olfactory bulb, are extensions of the brain's limbic system.

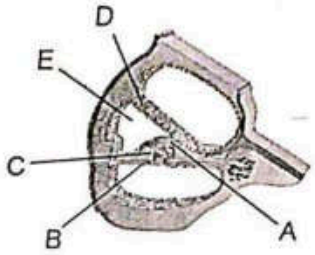
B : Cerebral cortex contains large regions that are neither clearly sensory nor motor in function and called as the association areas.

C : Hypothalamus contains number of centres which control body temperature, urge for eating and drinking.

D : Blind spot is a thinned-out portion of the retina where only the cones are densely packed.

	A	B	C	D
(1)	T	T	T	F
(2)	T	T	F	T
(3)	F	F	T	T
(4)	F	T	T	F

155. *Observe the given diagrammatic representation of the sectional view of cochlea.*



*Choose the option that represents the **correct** labelling.*

(1) A : *A thin elastic tympanic membrane*

(2) B : *Basilar membrane that separates endolymph filled scala media and perilymph filled scala tympani*

(3) E : *Auditory receptors containing Organ of Corti*

(4) D : *Membrane that separates perilymph filled scala media and endolymph filled scala vestibuli*

156. Approximate life span of crow is

(1) More than that of parrot

(2) Equivalent to that of fruit fly

(3) Less than that of crocodile

(4) More than that of tortoise

157. In humans __A__ ensures continuity of species between organisms of one generation and the next. Here 'A' is

(1) Ovum (2) Sperm

(3) Morula (4) Zygote

158. The female gamete undergoes development to form new organism without fertilization in all, **except**

(1) Rotifers (2) Honeybee

(3) Turkey (4) Earthworm

159. Complete the analogy w.r.t. chromosome number in meiocyte

Housefly : 12 : : Fruit fly : _____

(1) 6 (2) 8

(3) 24 (4) 20

160. Select the **odd one** w.r.t. post-fertilization event

(1) Insemination (2) Implantation

(3) Gestation (4) Parturition

161. How many seminiferous tubules approximately will be present in human testes?
(1) 500 –1500 (2) 250 – 400
(3) 125 – 250 (4) 100 – 200

162. Finger like projections at the edges of the funnel-shaped infundibulum are called
(1) Isthmus
(2) Oviduct
(3) Fimbriae
(4) Uterine fundus

163. Spermatids are transformed into sperms by the process called
(1) Spermiation
(2) Spermatogenesis
(3) Spermatocytogenesis
(4) Spermiogenesis

164. Read the following statements.
(a) Menarche refers to the first menstruation that begins at puberty.
(b) In a normal fertile female, menstruation only occurs if the released ovum is not fertilized.
(c) Both LH and FSH attain a peak level in the middle of menstrual cycle.
(d) hCG, hPL and relaxin are produced in pregnant women only.
Among the given statements, how many are **incorrect**?
(1) 1 (2) 2
(3) 3 (4) 0

165. Which among the following serve as contribution of scientists at CDRI, Lucknow?
(1) Saheli
(2) NACO
(3) RCH
(4) Nirodh

166. Select the factor that checks on population growth rate
(1) Decline in MMR
(2) Decline in IMR
(3) Decline in people in reproducible age
(4) Decline in death rate

167. Choose the **incorrect** match.
(1) *Periodic abstinence* - *Natural method of contraception*
(2) *Vaults, Lippes loop* - *Intrauterine devices*
(3) *Oral pills* - *Inhibits ovulation and implantation*
(4) *MTP* - *Voluntary termination of pregnancy*

168. Type of ART, in which zygote or early embryos with upto 8 blastomeres could be transferred into fallopian tube, is called
(1) GIFT (2) AI
(3) IUT (4) ZIFT

169. S.L. Miller created electric discharge in a closed flask containing which of the following conditions?
(1) CH_4, O_2, NH_3 at 800°C
(2) H_2, water vapour at 400°C
(3) CH_4, H_2, NH_3, water vapour at 800°C
(4) CO_2, H_2O, O_2 at 800°F

170. Select an example of Divergent evolution.
(1) Eye of the *Octopus* and of mammals
(2) Flippers of penguins and dolphins
(3) Wings of butterfly and birds
(4) Thorns and tendrils of *Bougainvillea* and *Cucurbita* respectively

171. Match column I with column II w.r.t. placental mammals and corresponding Australian marsupials.

Column I (Placental mammals)	Column II (Australian marsupials)
a. Lemur	(i) Numbat
b. Anteater	(ii) Spotted cuscus
c. Bobcat	(iii) Flying phalanger
d. Flying squirrel	(iv) Tasmanian tiger cat

Choose the **correct** option
(1) a-i, b-ii, c-iii, d-iv (2) a-ii, b-i, c-iv, d-iii
(3) a-ii, b-i, c-iii, d-iv (4) a-iv, b-iii, c-ii, d-i

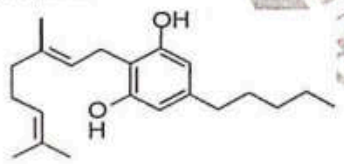
172. Land reptile that went back into water probably 200 mya to evolve into fish like reptiles called
(1) *Brachiosaurus* (2) *Ichthyosaurs*
(3) *Tyrannosaurus* (4) Coelacanth

173. Choose the **correct** answer w.r.t. bacterial disease only
(1) Dysentery, common cold, ascariasis
(2) Filariasis, typhoid, tuberculosis
(3) Dysentery, diphtheria, plague
(4) Amoebic dysentery, plague, typhoid

174. **Incorrect** match w.r.t. disease and causative agent is
(1) Malaria - *Plasmodium* sp.
(2) Elephantiasis - Female *Culex*
(3) Syphilis - *Treponema pallidum*
(4) Ringworms - *Microsporum*

175. Read the given statements.
(a) Bone marrow and thymus are primary lymphoid organs
(b) Colostrum provides artificial passive immunity to newborn.
(c) AIDS is not a congenital disease.
(d) Malignant cancer exhibit metastasis but not contact inhibition.
Choose the option representing only **incorrect** statements(s).
(1) (a), (c) and (d) (2) (b) and (d)
(3) (c) and (d) (4) (b) only

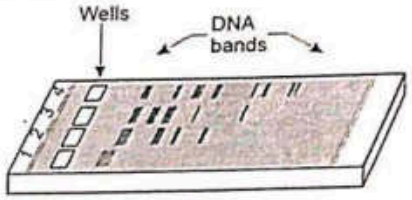
176. Identify the given structure and choose the **correct** option.



(1) Morphine
(2) Cannabinoids
(3) Smack
(4) Coke

177. *Hisardale* is a result of cross between
(1) Bikaneri ewes and Merino rams
(2) Merino ewes and Bikaneri rams
(3) Brown Swiss and Sahiwal
(4) Brown Swiss and Bikaneri ewes.

178. Based on the given figure representing agarose gel electrophoresis, choose the **correct** statement.



(1) Lane 1 shows migration of digested DNA
(2) Lane 4 shows DNA fragmentation of different species
(3) Smallest DNA fragment moves farthest from cathode under sieving effect
(4) Largest DNA moves fastest and present closest to anode attributing to its negative charge

179. If gene of interest A and B are inserted in *Bam* HI and *Pvu* I site in pBR322, the transformed recombinants will
(1) Be ampicillin sensitive and tetracycline resistance
(2) Not grow on antibiotic containing medium
(3) Be tetracycline and ampicillin resistant
(4) Be of blue color with tetracycline sensitivity

180. Select the **incorrect** match.
(1) Tracy - Transgenic sheep
(2) ANDi - Transgenic monkey
(3) Dolly - Transgenic sheep
(4) Rosie - Transgenic cow